A Portable Dynamic Student Dropout Early Warning Model

MBAI 5600G

Applied Integrative Analytics Capstone Project - Final Report

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Abstract

This project develops one fixed student-risk model that can accept different local column names through a documented semantic adapter. The main model uses 16 numeric features: four required background and timing fields, five optional background fields, four assessment fields, and three platform-neutral activity fields. UCI contributes enrolment snapshots, while OULAD contributes enrolment and day-35, day-60, and day-75 snapshots. One pooled XGBoost model is then used without changing its feature order or fitted parameters. On held-out data, ROC-AUC was 0.771 for UCI enrolment and 0.595, 0.677, 0.694, and 0.689 for the four OULAD snapshots. OULAD-only models were better at the three later snapshots by 0.009 to 0.013 ROC-AUC. These small differences were statistically significant for ROC-AUC but not for average precision. A separate OULAD-specific 51-feature model reached 0.717 ROC-AUC at day 35, showing the performance cost of removing platform-specific detail. The result is a reusable model and mapping process, not proof of universal transfer. A third institution is still needed for external validation of the dynamic fields.

Keywords: student dropout; early warning; model portability; semantic adapter; XGBoost; learning analytics

1. Introduction

Student withdrawal is not only a final academic outcome. It is usually preceded by missing work, low assessment results, or reduced participation. The useful business question is whether an institution can identify a manageable group for support while a student is still registered. A model that uses final grades may predict well but arrive too late for an advisor to act.

Portability creates a second problem. A campus program and an online university do not store the same columns. UCI contains admissions, finance, and semester records. OULAD contains registrations, dated assessments, and learning-platform events [3], [4]. A model tied to either raw table cannot be moved directly to another school. This project treats field mapping as part of the model design instead of assuming that equal column names exist.

The final objective is one versioned XGBoost model with a fixed semantic input contract. Each school keeps its local systems and provides an adapter that calculates the required concepts. Optional feeds can be blank. The output is a local ranking score for human review, not an automatic dropout decision or a calibrated personal probability.

2. Problem Statement

The project asks whether one early-warning model can score students from different institutions even when their source columns do not match. The technical challenge is to define features with stable meanings, protect prediction timing, keep one student out of more than one data split, and measure the performance lost when institution-specific detail is removed.

The research questions are:

RQ1. Can the UCI experiment from the base paper be reproduced closely enough to provide a benchmark?

RQ2. Can UCI and OULAD records be converted to one fixed semantic schema and scored by the same fitted model?

RQ3. How does the pooled model compare with local-only models at enrolment and later OULAD snapshots?

RQ4. Which features become important as dated assessment and activity information becomes available?

RQ5. What checks would a third school need before using the scores in an advising workflow?

The work is retrospective and uses public, de-identified data. It does not include Ontario Tech student records or a live intervention. All experiments ran on a Mac laptop. This limited the model search to Logistic Regression and a small set of XGBoost settings, but it did not require a record-count restriction.

3. Background & Literature Review

Islam et al. [1] combined academic prediction with explainable AI and a proposed support system. Their UCI experiment was selected as the base paper. Realinho et al. [2] explain that the UCI dataset contains both enrolment variables and later semester outcomes. This timing matters because first- and second-semester grades are not day-one predictors. Our reproduction kept the three-class full-feature task for comparison, while the portable models use only information appropriate to their stated snapshot.

OULAD provides 32,593 student-module records and more than ten million learning-platform events [4]. Prior work shows that assessment and engagement history can support early prediction [6], [13]. However, reported performance depends on the prediction date, eligible cohort, outcome definition, and split method. A row-level random split is especially risky because one OULAD student can appear in more than one module.

The literature contains many stronger single-dataset models, so the contribution here is not a new classifier. It is the portable measurement layer, one actual model object used across snapshots, structured tolerance for unavailable feeds, paired uncertainty tests, and a working new-school scoring example. SHAP is used to describe the fitted model [8], but it is not interpreted as a causal explanation. TRIPOD+AI and PROBAST+AI were used as reporting and risk-of-bias guides rather than educational standards [9], [10].

4. Data Description

Table 1 summarizes the two data sources and their roles. UCI supplies 4,424 student records and a binary dropout target for portability tests. OULAD supplies dated registration, assessment, and activity history. The final result Withdrawn is positive only when the recorded unregistration occurs after the snapshot. A student who had already withdrawn is not left in a later risk set.

Table 1. Data sources and analytical roles

Dataset	Setting	Main size	Role	Positive outcome
UCI	Portuguese higher education	4,424 students	Base-paper benchmark and enrolment snapshots	Dropout
OULAD	UK distance learning	32,593 student-module records; >10M events	Enrolment and day 35/60/75 snapshots	Future withdrawal after snapshot

Table 1 also shows that the endpoints are related but not identical. UCI predicts final student status, while OULAD predicts a future withdrawal from an individual module presentation. This difference is retained as a limitation rather than hidden by relabeling both outcomes.

Table 2 reports the changing OULAD risk sets. Later cohorts are smaller because students who already withdrew are removed. Prevalence falls from 0.237 at enrolment to 0.134 by day 75. Average precision must therefore be read beside prevalence rather than compared as if every row came from the same class balance.

Table 2. OULAD snapshot cohorts before model evaluation

Snapshot	Rows	Unique students	Future-withdrawal prevalence
Day 0	29,161	26,085	0.237
Day 35	27,129	24,567	0.174
Day 60	26,253	23,899	0.146
Day 75	25,795	23,566	0.131

4.1 Methodology

Figure 1 shows the end-to-end design. Raw fields are not passed directly to the model. A local adapter converts them to the same 16 numeric concepts. The frozen pipeline then applies its fitted imputer and XGBoost model. At a new school, scores are converted to percentiles within the same course and snapshot day, or within another clearly defined operational cohort, so advisor capacity can determine the review list.

Figure 1. Portable model training and new-school scoring workflow

Local raw data > **Semantic adapter** > **16 fixed features** > **Frozen XGBoost** > **Local risk ranking**

Figure 1 is also the answer to the different-column problem. The model does not discover an unknown field automatically. A school must document what a local field means, its unit, its timing, and how it maps to the contract. The supplied JSON example handles local names and scales without refitting the model. Invalid required values, binary fields, and course lengths are rejected.

Four features are required: scaled age, prior education level, study load, and course progress ratio. Twelve fields are optional. During training, entire optional background, assessment, and activity groups were randomly masked. This lets the same column order remain valid when a school lacks one feed. Median imputation is fitted on training data only.

Students were assigned to 60% training, 20% validation, and 20% test partitions. Every module and snapshot for one student followed the same assignment. Split-overlap checks returned zero. Training weights first prevented students with several module snapshots from dominating. The four institution-by-outcome cells then received equal total weight. This keeps both institutions and both classes represented in the pooled objective.

4.2 Exploratory Data Analysis

Figure 2 shows the original UCI target distribution. Graduate is the largest class, followed by Dropout and Enrolled. The portable binary task combines Enrolled and Graduate as the non-dropout class. This creates some label uncertainty because Enrolled is not a final successful outcome.

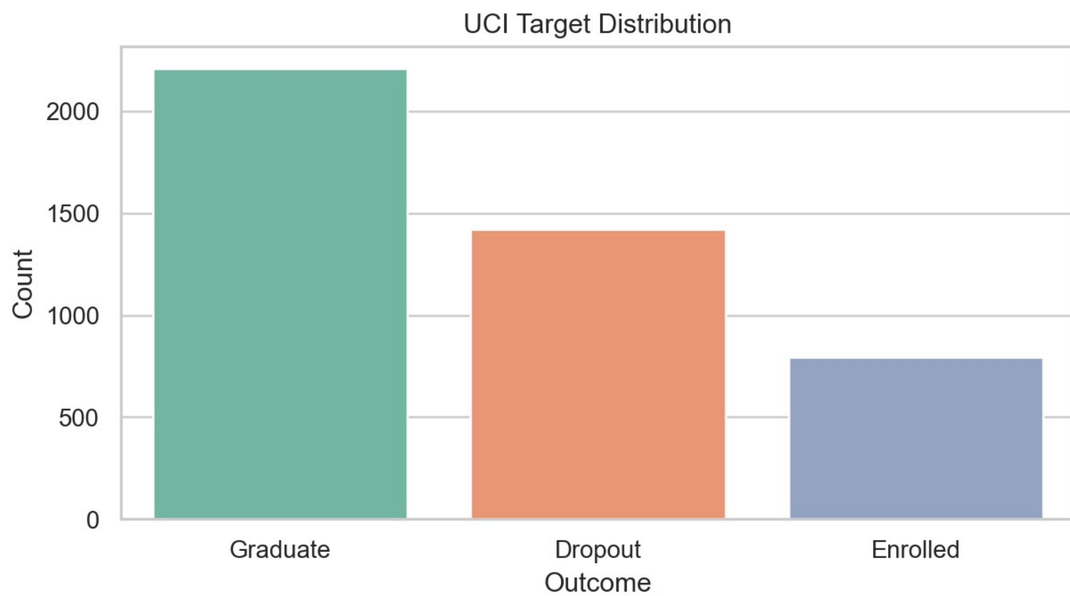

Figure 2. UCI outcome distribution

Figure 3 shows the four OULAD final-result categories. Pass is the largest group. Withdrawn is smaller than the combined Pass and Distinction groups, which is why accuracy alone would be misleading. The operational target is narrower again because only future withdrawals among currently registered students count as positives.

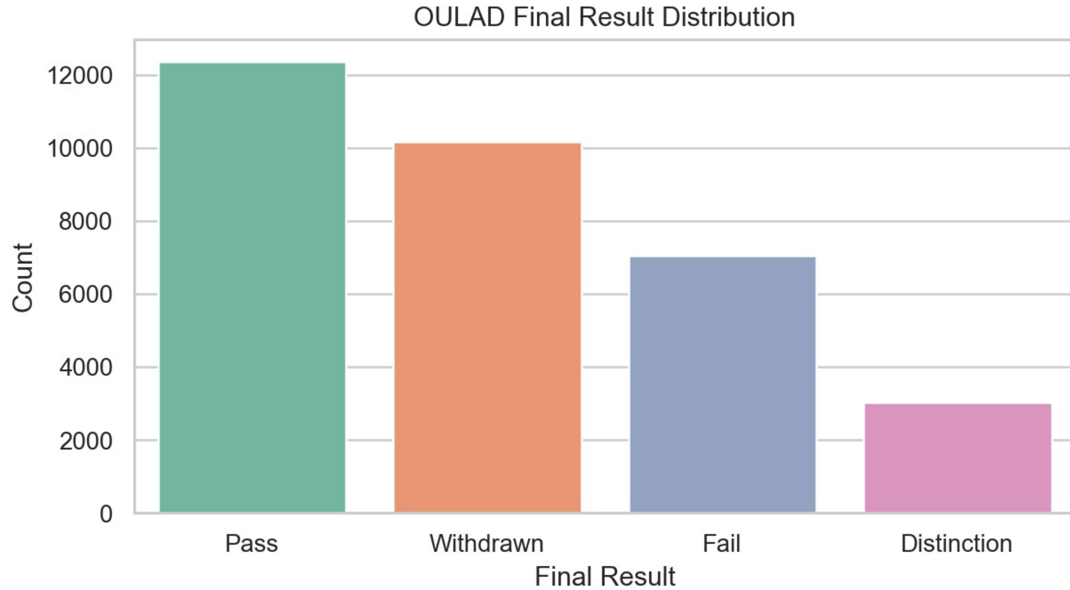

Figure 3. OULAD final-result distribution

Figure 4 compares total learning-platform activity by final result. Withdrawn students generally have lower activity, but the distributions overlap. A low activity value is therefore a warning signal rather than proof that a student will leave. This pattern motivated platform-neutral active-day and recency features instead of raw click-type columns.

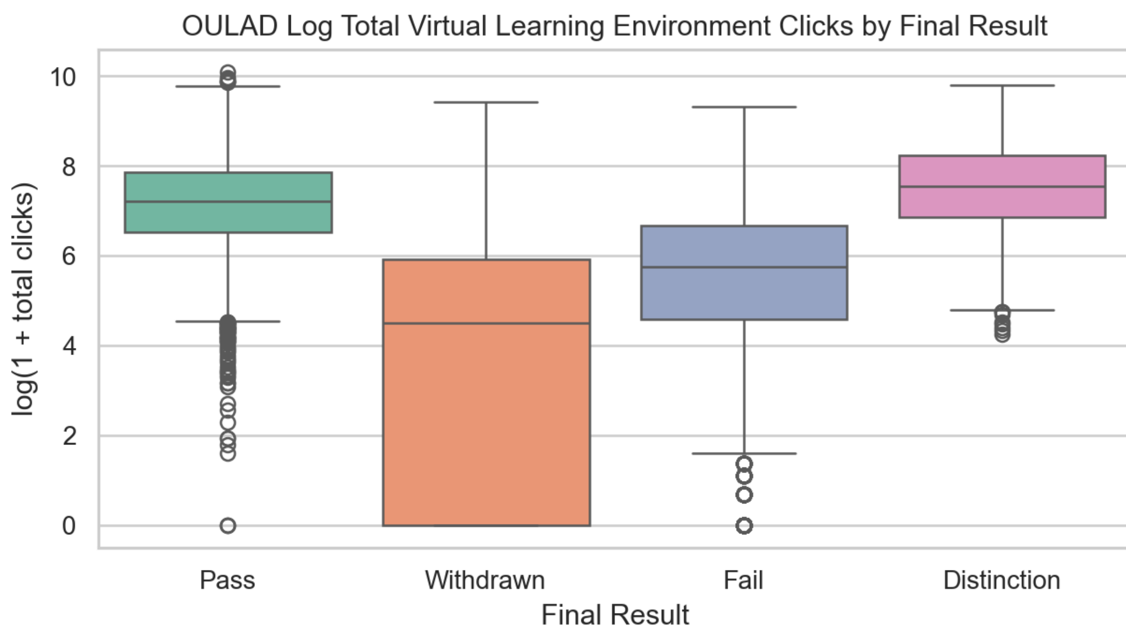

Figure 4. OULAD learning-platform activity by final result

4.3 Data Limitations

UCI has no dated learning-platform or assessment events, so it contributes only enrolment snapshots to the dynamic model. OULAD is the only source for the later optional fields. As a result, the project demonstrates that one model object can accept missing or available feature groups, but it does not externally validate dynamic fields across two schools. A third institution with dated events is required for that claim.

Several semantic mappings are approximate. Financial stability is based on debt and tuition status in UCI and an area-deprivation midpoint in OULAD. Study load also uses institution-specific heavy-load denominators. These values share a documented direction and range but are not equal instruments.

5. AI/ML Models

Logistic Regression was retained as a linear baseline and XGBoost was used for the final nonlinear model [7]. Table 3 lists the fixed settings for the 16-feature experiment. The settings were selected from earlier milestone work and then held constant across local, zero-shot, and pooled scenarios. Thresholds for F1 were selected on validation data separately for each snapshot. Operational scoring does not reuse these thresholds at a new school; it returns rankings instead.

Table 3. Main model and preprocessing settings

Component	Setting	Reason
Missing values	Training median; empty columns retained	One fixed schema with optional feeds
Logistic Regression	C=0.2; standardized inputs; 3,000 iterations	Linear baseline
XGBoost	320 trees; depth 5; learning rate 0.03	Nonlinear tabular model
Sampling	subsample=0.85; colsample=0.85	Reduce dependence on individual rows and features
Regularization	min child weight=4; L2=2.0; L1=0.1	Control model complexity
Random seed	42	Reproducible split and training

The report uses accuracy only for the base-paper reproduction. For binary results, precision, recall, F1, ROC-AUC, and average precision (AP) are reported. Let TP, TN, FP, and FN denote the four confusion-matrix counts, y_i the observed outcome, p_i a model score, and n the number of records.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) \tag{1}$$

$$\text{Precision} = TP / (TP + FP) \tag{2}$$

$$\text{Recall} = TP / (TP + FN) \tag{3}$$

$$F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall}) \tag{4}$$

$$\text{ROC-AUC} = P(\text{score}_{positive} > \text{score}_{negative}) \tag{5}$$

$$AP = \sum_k [(\text{Recall}_k - \text{Recall}_{(k-1)}) \times \text{Precision}_k] \tag{6}$$

$$\text{Brier} = (1/n) \times \sum_i (p_i - y_i)^2 \tag{7}$$

Equations (1)-(4) describe threshold-based performance. Equation (5) measures ranking across positive and negative records. Equation (6) summarizes precision across recall levels and is useful under class imbalance. Equation (7) measures squared probability error, although the final deployment score is explicitly treated as uncalibrated.

Model comparisons used 1,000 student-cluster bootstrap samples and 1,000 paired student-cluster permutations. All records for one student were resampled or swapped together. The two-sided Monte Carlo p-value was calculated by Equation (8), where B is the number of permutations.

$$p = [1 + \sum_b I(|d_b| >= |d_{observed}|)] / (B + 1) \tag{8}$$

SHAP summaries use mean absolute contribution, defined in Equation (9). This quantity describes how much a feature moves the fitted model output on average; it does not show that changing the feature would cause retention.

$$\text{Mean absolute SHAP}_j = (1/n) \times \sum_i |\text{SHAP}_{ij}| \tag{9}$$

6. Experimental Results and Analysis

6.1 Main Portable Dynamic Model

Table 4 contains the main held-out results for one pooled 16-feature XGBoost model. UCI enrolment ROC-AUC was 0.771. OULAD improved from 0.595 at enrolment to 0.677 at day 35 and 0.694 at day 60, then remained similar at 0.689 on day 75. AP falls after day 35 partly because the future-withdrawal prevalence also falls.

Table 4. One pooled dynamic model across held-out contexts

Held-out context	Rows	Prevalence	ROC-AUC	AP	Precision	Recall	F1
OULAD enrolment	5,811	0.237	0.595	0.310	0.237	1.000	0.384
OULAD day 35	5,408	0.175	0.677	0.311	0.309	0.450	0.367
OULAD day 60	5,240	0.148	0.694	0.292	0.279	0.504	0.359
OULAD day 75	5,154	0.134	0.689	0.270	0.241	0.485	0.322
UCI enrolment	885	0.321	0.771	0.635	0.469	0.827	0.599

Figure 5 visualizes the same ranking and threshold metrics. The model is useful as a moderate ranking system, but its raw score should not be presented as a personal dropout probability. The day-0 OULAD F1 reflects a validation threshold that classified nearly every positive but also many negatives, which is another reason to use a locally capacity-based review rule.

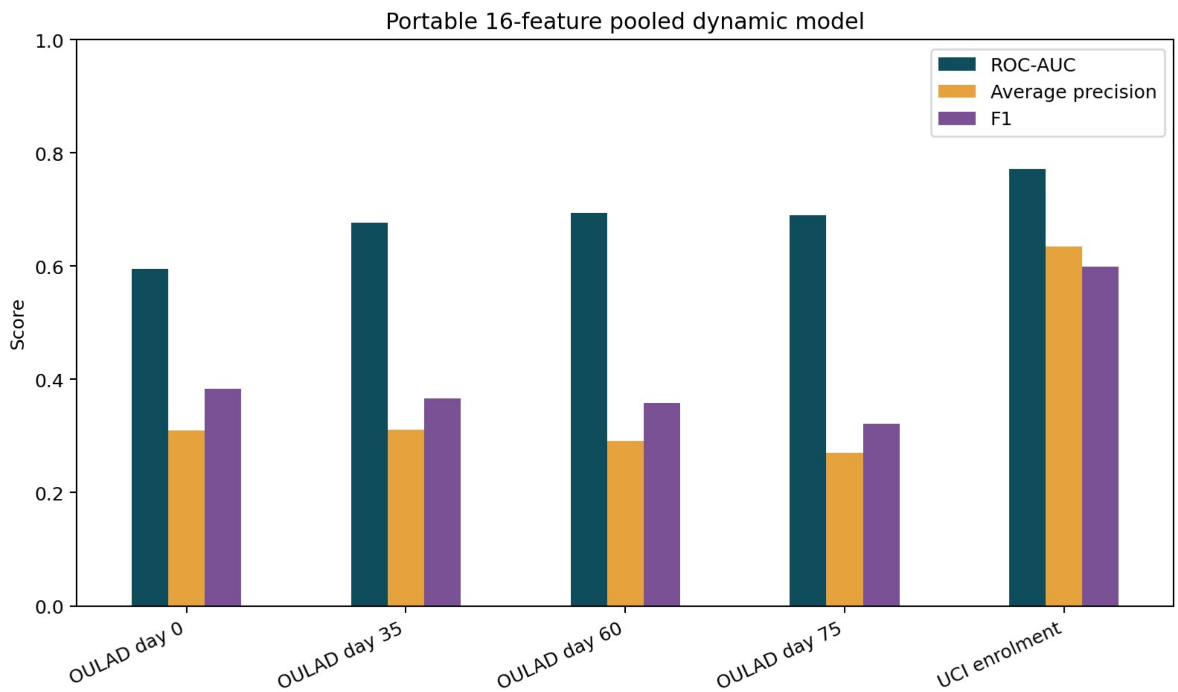

Figure 5. Performance of one pooled 16-feature dynamic model

6.2 Portability Cost and Statistical Testing

Table 5 compares the pooled XGBoost with the corresponding local-only XGBoost on exactly the same held-out records. The UCI and OULAD enrolment differences were not significant. At OULAD days 35, 60, and 75, pooled ROC-AUC was lower by 0.009, 0.013, and 0.009. The paired tests were significant, although the effect sizes were small. AP differences at those snapshots did not reach 0.05.

Table 5. Paired student-cluster ROC-AUC tests for pooled minus local-only XGBoost

Comparison	Difference	95% cluster CI	p-value	Significant
UCI day 0: pooled minus local-only	-0.003	[-0.011, +0.007]	0.561	No
OULAD day 0: pooled minus local-only	-0.001	[-0.008, +0.006]	0.740	No
OULAD day 35: pooled minus local-only	-0.009	[-0.015, -0.003]	0.005	Yes
OULAD day 60: pooled minus local-only	-0.013	[-0.018, -0.007]	0.001	Yes
OULAD day 75: pooled minus local-only	-0.009	[-0.015, -0.003]	0.005	Yes

Table 5 changes the conclusion from the older draft. The unified model is not statistically equal to the OULAD-only model at later snapshots. Its advantage is operational reuse with a measured performance cost, not higher accuracy.

Table 6 gives a descriptive comparison with the OULAD-specific 51-feature benchmark. The specialized model has higher ROC-AUC at every later snapshot and reaches 0.717 at day 35. It uses assignment and platform detail that the portable schema intentionally removes. The rows are not used for a paired significance claim because the two analyses do not have perfectly identical test cohorts.

Table 6. Portable dynamic model and OULAD-specific benchmark

Model scope	Day	ROC-AUC	AP	F1
Portable 16	35	0.677	0.311	0.367
Portable 16	60	0.694	0.292	0.359
Portable 16	75	0.689	0.270	0.322
OULAD-specific 51	35	0.717	0.356	0.403
OULAD-specific 51	60	0.718	0.305	0.379
OULAD-specific 51	75	0.717	0.277	0.343

6.3 Explanation Across Snapshots

Figure 6 summarizes SHAP values from the same pooled model. Financial stability and age contribute more for UCI enrolment. OULAD day 35 places more weight on assessment average score. Course progress ratio becomes large at later OULAD snapshots because the model is shared across time. Within one fixed snapshot this value is constant, so it shifts the model baseline but cannot rank students inside that day by itself. It should not be turned into an intervention.

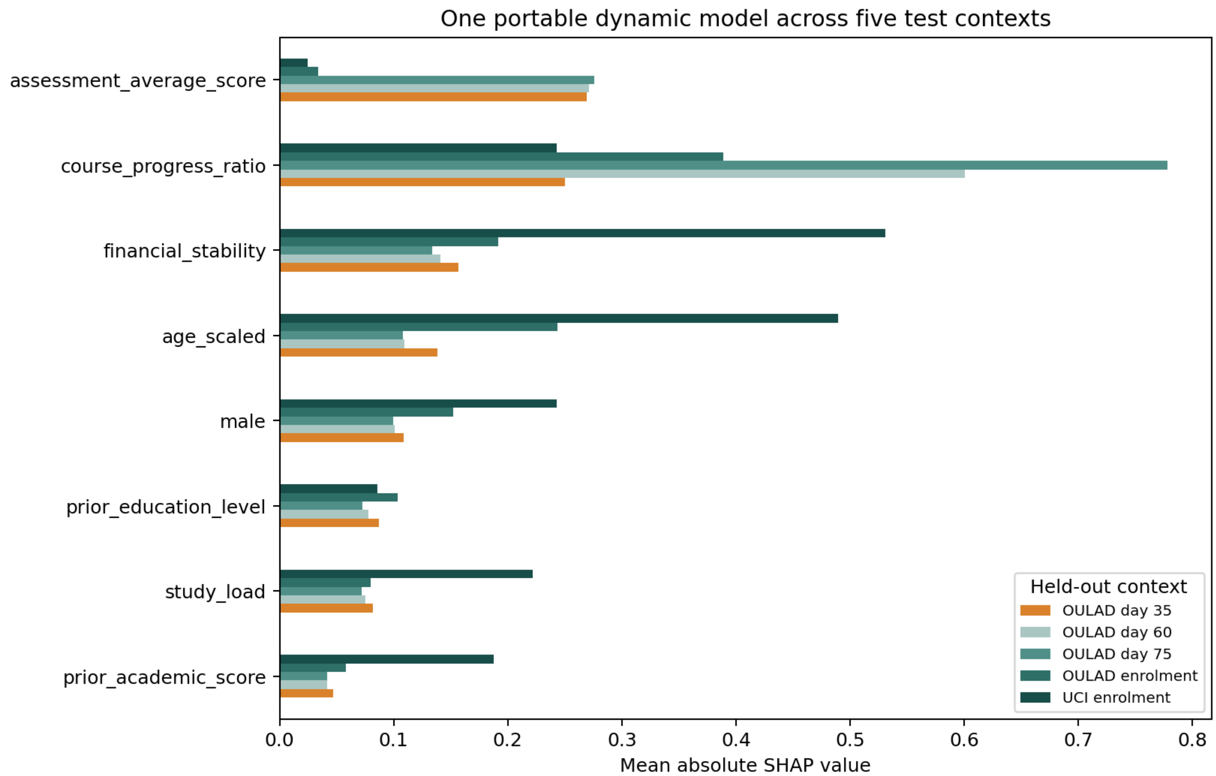

Figure 6. Mean absolute SHAP for one model in five held-out contexts

The feature shifts in Figure 6 support local interpretation. An advisor action should depend on the available local evidence. A low assessment score may lead to academic outreach, while missing activity may first require checking access, work, or technical barriers. Sensitive optional fields should not be used as the reason for an adverse action.

6.4 Enrolment-Only Baseline and Supporting Studies

The six-feature enrolment model is retained as a fallback for schools without dated events. Table 7 shows that pooled ROC-AUC was 0.777 on UCI and 0.624 on OULAD. Direct UCI-to-OULAD transfer was only 0.481. This supports pooled training, but it also shows that shared feature names do not remove institutional shift.

Table 7. Six-feature enrolment model results

Experiment	ROC-AUC	AP
UCI local -> UCI	0.777	0.644
OULAD local -> OULAD	0.635	0.418
UCI -> OULAD zero-shot	0.481	0.298
OULAD -> UCI zero-shot	0.603	0.407
Pooled -> UCI	0.777	0.635
Pooled -> OULAD	0.624	0.409

Figure 7 displays the enrolment transfer experiments and the earlier local adaptation analysis. It is supporting evidence rather than the final dynamic design. The main lesson is that a small target-school sample can expose a weak zero-shot model before operational use.

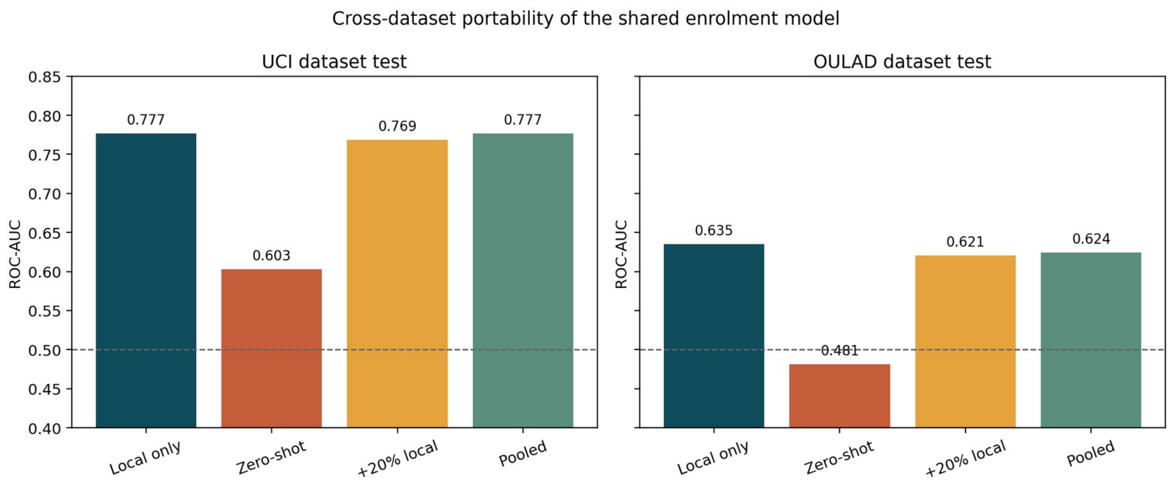

Figure 7. Six-feature local, zero-shot, adapted, and pooled transfer results

The base-paper reproduction reached 0.768 accuracy and 0.704 macro F1 on the UCI three-class task, below the 0.83 accuracy reported by Islam et al. [1]. The original full protocol was not available, so this is treated as a close reproduction. In the stricter UCI day-one binary benchmark, XGBoost reached ROC-AUC 0.832 and F1 0.666. The OULAD-specific analysis reached 0.717 ROC-AUC at each of days 35, 60, and 75, and a later 2014J presentation holdout dropped to 0.678. These results show temporal and setting shift even inside one dataset.

Figure 8 shows the OULAD-specific temporal results. ROC-AUC stayed stable while AP declined in the changing operational cohorts. In a controlled day-75 cohort, later feature windows improved ranking. This separates added information from the changing population of students still registered at each date.

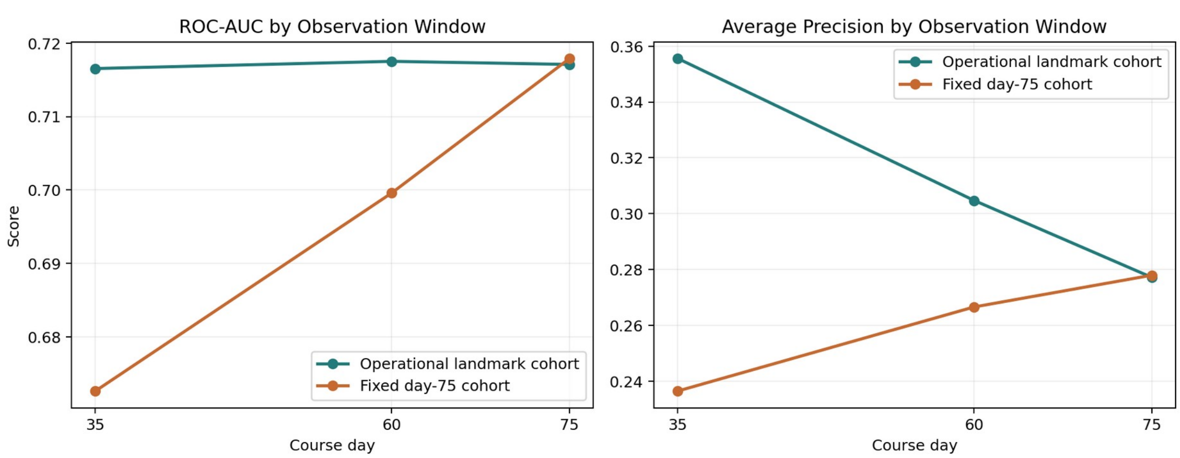

Figure 8. OULAD-specific temporal performance and controlled-cohort comparison

7. Business Impact

The proposed output is an advisor review queue. A school can select a review share that matches staffing, such as the top 15% within each course at the same snapshot day. For 1,000 active enrolments, this produces 150 records for review. It does not claim that 150 students will withdraw or that all should receive the same action.

Table 8 converts the model into a cautious pilot. The first run should be silent and historical. Staff should verify the adapter, outcome timing, ranking metrics, subgroup performance, and score stability before alerts are shown. The model and adapter need separate version numbers because a local system change can break a mapping without changing the classifier.

Table 8. Minimum third-school pilot gates

Gate	Evidence	Decision
Semantic mapping	Document source, unit, range, and availability date	Reject unclear mappings
Technical validation	No missing required fields; expected optional coverage	Repair feed before scoring
Historical shadow test	ROC-AUC, AP, calibration, and capacity precision	Do not use weak rankings
Fairness review	Error and score checks by approved subgroups	Restrict or revise sensitive features
Operational test	Advisor capacity, response time, and referral options	Set local review share
Monitoring	Drift, mapping changes, and outcomes by term	Pause when contract changes

Table 8 is important because the present study does not estimate financial savings or causal retention gains. The business value at this stage is a repeatable way to prioritize review and a smaller integration task than rebuilding a model around every local table. A later pilot would need to measure whether the support action itself improves persistence [14].

8. Ethical and Responsible AI

Student data remain sensitive after names are removed. Access should be limited to staff with a support role, and the school should define retention periods, audit logs, and an appeal or correction route. Scores should not be used for admission, discipline, financial penalties, or reduced services. The NIST AI RMF provides a useful structure for ownership and monitoring [11].

Age, gender, disability, financial variables, and their proxies may reproduce existing inequality. Gender and declared support need are optional in the contract. A school should compare the full model with a restricted version and review error rates and score distributions for relevant groups. A SHAP value does not prove fairness and does not explain a student's personal reason for leaving.

Human review remains necessary. An alert should start a supportive conversation and a data-quality check. It should not be presented as a fact about the student. Advisors also need realistic services to offer; otherwise, a larger alert list does not create value.

9. Limitations and Future Work

Only two public datasets were used, and only OULAD has dated dynamic fields.

UCI dropout and OULAD module withdrawal are related but different outcomes.

The semantic mappings are documented approximations rather than validated measurement equivalence.

The pooled model is slightly but significantly worse than OULAD-only models at later snapshots.

Thresholds and raw scores are not calibrated for a third institution.

The study is retrospective and does not show that advisor outreach causes retention.

The next step is a pre-registered third-school test. The 16-feature contract and saved model should be frozen before viewing the new outcomes. The school should first map an historical cohort, report feature coverage and drift, and evaluate the unchanged model. If the ranking is weak, limited local adaptation or a hierarchical model can be tested without touching the original external test. A later prospective study should compare support outcomes for reviewed and non-reviewed students.

10. Conclusions

The project produced one deployable model object, a 16-feature semantic contract, a configurable mapping script, and a scoring script that works with missing optional feeds. The pooled model reached ROC-AUC 0.771 on UCI enrolment and 0.677 to 0.694 on later OULAD snapshots. Local OULAD models remained slightly better, and the OULAD-specific 51-feature model was stronger again. Portability therefore has a measurable cost.

The practical outcome is not a universal plug-and-play predictor. It is a reusable core model with local measurement, local validation, ranking-based use, and human governance. The design can be applied to different column names, but a third school must still prove that its fields carry the intended meanings and that the resulting queue is safe and useful.

11. References

[1] M. M. Islam, F. H. Sojib, M. F. H. Mihad, M. Hasan, and M. Rahman, "The integration of explainable AI in Educational Data Mining for student academic performance prediction and support system," Telematics and Informatics Reports, vol. 18, Art. no. 100203, 2025.

[2] V. Realinho, J. Machado, L. Baptista, and M. V. Martins, "Predicting Student Dropout and Academic Success," Data, vol. 7, no. 11, Art. no. 146, 2022.

[3] V. Realinho et al., "Predict Students' Dropout and Academic Success," UCI Machine Learning Repository, 2021. doi: 10.24432/C5MC89.

[4] J. Kuzilek, M. Hlosta, and Z. Zdrahal, "Open University Learning Analytics dataset," Scientific Data, vol. 4, Art. no. 170171, 2017.

[5] B. Carballo-Mendivil et al., "Predicting Student Dropout from Day One: XGBoost-Based Early Warning System Using Pre-Enrollment Data," Applied Sciences, vol. 15, no. 16, Art. no. 9202, 2025.

[6] A. Shaikhanova et al., "Interpretable Predictive Modeling for Educational Equity," Big Data and Cognitive Computing, vol. 9, no. 11, Art. no. 297, 2025.

[7] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. ACM SIGKDD, 2016, pp. 785-794.

[8] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Advances in Neural Information Processing Systems, vol. 30, 2017.

[9] G. S. Collins et al., "TRIPOD+AI statement," BMJ, vol. 385, e078378, 2024.

[10] K. G. M. Moons et al., "PROBAST+AI," BMJ, vol. 388, e082505, 2025.

[11] National Institute of Standards and Technology, Artificial Intelligence Risk Management Framework (AI RMF 1.0), NIST AI 100-1, 2023.

[12] C. Romero and S. Ventura, "Educational data mining and learning analytics: An updated survey," WIREs Data Mining and Knowledge Discovery, vol. 10, no. 3, e1355, 2020.

[13] F. Qiu et al., "Predicting students' performance in e-learning using learning process and behaviour data," Scientific Reports, vol. 12, Art. no. 453, 2022.

[14] S. Plak et al., "Early warning systems for more effective student counselling in higher education," Higher Education Quarterly, vol. 76, no. 1, pp. 131-152, 2022.

Appendix A. Portable Dynamic Feature Contract

Table A1 lists the fixed model inputs. Required fields must be present for every scored record. Optional fields may be absent or blank and are imputed by the frozen pipeline. All observed values are on documented 0-1 scales.

Table A1. Sixteen-feature semantic contract

Feature	Status	Group	Meaning
age_scaled	Required	background	Age position on a fixed 18-60 scale
prior_education_level	Required	background	Highest prior education on a documented 0-1 ladder
study_load	Required	background	Registered load relative to a heavy full load
course_progress_ratio	Required	timing	Share of planned course duration elapsed
prior_academic_score	Optional	background	Normalized pre-entry grade or GPA
previous_attempts	Optional	background	Prior attempts capped and scaled to 0-1
male	Optional	sensitive_optional	Optional documented binary gender indicator
declared_support_need	Optional	sensitive_optional	Optional declared disability or support need
financial_stability	Optional	sensitive_optional	Optional documented financial-stability proxy
assessment_completion_rate	Optional	assessment	Share of due assessment weight submitted by snapshot
assessment_average_score	Optional	assessment	Weighted normalized score among submitted assessments
assessment_score_available	Optional	assessment	Whether at least one score is available at snapshot
late_submission_rate	Optional	assessment	Share of submitted assessments submitted after due date
active_day_rate	Optional	activity	Share of observable course days with LMS activity
days_since_last_activity_scaled	Optional	activity	Days since last LMS activity capped at 30 and divided by 30
recent_activity_rate	Optional	activity	Share of observable days active in the last 14 days

Table A1 is implemented in the JSON contract and mapping code. Raw course identifiers, regions, assignment numbers, and platform activity types are excluded because their meanings do not transfer reliably.

Appendix B. Reproducibility Checklist

Table B1. Submitted code and data package

Item	Location	Purpose
Main model	code/cross_school_dynamic_model.py	Build snapshots, train, test, explain, and save model
New-school mapper	code/map_new_school_snapshots.py	Convert local columns and units
Scoring	code/score_new_school_dynamic.py	Create ranking, percentile, and review flag
Output checks	code/check_outputs.py	Check schemas, splits, artifacts, metrics, and examples
Frozen artifact	models/unified_dynamic_xgboost.joblib	One fitted pipeline for scoring
Contract	models/unified_dynamic_feature_contract.json	Feature definitions and inference policy
Raw data	data/raw	Packaged UCI and OULAD sources
Results	outputs/results	Metrics, predictions, tests, SHAP, and examples

Table B1 identifies the main reproducibility files. The README provides Windows, macOS, and Linux commands. The random seed is 42, exact dependencies are listed in requirements.txt, and check_outputs.py must finish with 'Output check passed.'